

Conservation of Eyes with Choroidal Melanoma by a Multimodality Approach to Treatment

An Audit of 1632 Patients

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Objective: To report on conservation of eyes with choroidal melanoma with a multimodality approach to treatment.

Design: Prospective, noncomparative, interventional case series.

Participants: All 1632 patients with choroidal melanoma treated at a single center between 1993 and 2002.

Intervention: Primary enucleation (35%), brachytherapy (31.3%), proton beam radiotherapy (16.7%), transscleral local resection (11.0%), endoresection (3.7%), transpupillary thermotherapy (2.5%), and photocoagulation (0.1%).

Main Outcome Measures: Primary and secondary enucleation.

Results: Logistic regression showed the main predictive factors for primary enucleation to be: age more than 60 years (odds ratio [OR], 2.4; 95% confidence interval [CI], 1.8%–3.2%); reduced visual acuity (OR, 2.5; 95% CI, 1.9%–3.2%); posterior extension close to or involving the optic disc and fovea (OR, 1.7; 95% CI, 1.2%–2.4%); circumferential spread around the ciliary body, iris, or angle (OR, 3.1; 95% CI, 1.8%–5.5%); basal tumor diameter (OR, 3.5; 95% CI, 2.4%–5.0%); and tumor height (OR, 6.3; 95% CI, 4.5%–8.9%). After conservative treatment, the actuarial rate of secondary enucleation was 11.1% at 5 years (95% CI, 8.6%–13.6%). Cox multivariate analysis indicated the factors independently predicting secondary enucleation as: nasal/midline tumor location (risk ratio [RR], 2.6; 95% CI, 1.6%–4.4%); disc involvement (RR, 2.2; 95% CI, 1.2%–4.1%); tumor diameter (RR, 1.2; 95% CI, 1.0%–1.5%); and tumor thickness (RR, 1.8; 95% CI, 1.5%–2.1%).

Conclusions: With a multimodality approach to treatment, 65% of patients underwent conservative treatment, and of them, 89% retained their eye at 5 years, with success depending on tumor diameter, tumor thickness, disc involvement, and coronal location. *Ophthalmology* 2004;111:977–983 © 2004 by the American Academy of Ophthalmology.

Uveal melanoma is the most common primary intraocular malignancy, with an incidence of approximately 5 per 1 000 000 per year.¹ The prime objective of treating the primary tumor is to reduce the threat of metastatic disease as much as possible. For many years, the only way of achieving this aim was by enucleation, and even today this is the only treatment that can be undertaken by most ophthalmologists. During the past few decades, several techniques have been developed that are aimed at conserving the eye with as

much vision as possible.² Such conservative methods include plaque radiotherapy,^{3,4} proton beam radiotherapy,⁵ stereotactic radiotherapy,⁶ photocoagulation,⁷ transpupillary thermotherapy,⁸ transscleral local resection,⁹ and transretinal endoresection.¹⁰

Most ocular oncology centers have 1 or 2 forms of treatment at their disposal. Some have facilities for a wide range of treatments and therefore are able to adopt a multimodality approach, selecting the most appropriate therapy, or combination of therapies, according to the size and location of the tumor as well as the patient's preferences.

The published literature on the treatment of uveal melanoma consists mostly of case series receiving 1 kind of treatment, or comparisons between 2 different methods. These studies usually do not describe the outcomes in patients not meeting the inclusion criteria. Such investigations are useful for defining the scope of each treatment and perhaps determining which of 2 rival treatments should be selected for a particular tumor; however, they do not give

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